Problem Statement Question 1

A biomechanics researcher analyzes a 3D musculoskeletal region where the **muscle force density** is given by:

$$g(x, y, z) = 652.75x^4y^8z^7$$

over the region bounded by: $0 \leq x \leq 5$, $0 \leq y \leq 6$, $0 \leq z \leq 3$. Compute the total muscle force distributed in the region.

Solution

The total muscle force F is defined as the triple integral of the density function $g(x, y, z)$ over the rectangular region V:

$$F = \iiint_V g(x, y, z)\, dV$$

Given the bounds of the region, the limits of integration are:

$$0 \leq x \leq 5, \quad 0 \leq y \leq 6, \quad 0 \leq z \leq 3$$

Applying the iterated integration process step-by-step:

$$F = \int_0^3 \int_0^6 \int_0^5 652.75x^4y^8z^7\, dx\, dy\, dz$$

Step 1: Integrate with respect to x.

$$\int_0^5 652.75x^4y^8z^7\, dx = 652.75y^8z^7 \left[\frac{x^5}{5}\right]_0^5$$

$$= 652.75y^8z^7 \left(\frac{3125}{5}\right) = 652.75y^8z^7(625) = 407968.75y^8z^7$$

$$\Rightarrow F = \int_0^3 \int_0^6 407968.75y^8z^7\, dy\, dz$$

Step 2: Integrate with respect to y.

Next, we integrate the resulting expression with respect to y:

$$\int_0^6 407968.75y^8z^7\, dy = 407968.75z^7 \left[\frac{y^9}{9}\right]_0^6$$

Substituting the bounds gives:

$$= 407968.75z^7 \left(\frac{10077696}{9}\right)$$

$$= 407968.75z^7(1119744) = 456820500000z^7$$

The integral expression simplifies to:

$$F = \int_0^3 456820500000z^7 \, dz$$

Step 3: Integrate with respect to z.

Finally, we evaluate the outermost integral with respect to z:

$$\int_0^3 456820500000z^7 \, dz = 456820500000 \left[\frac{z^8}{8}\right]_0^3$$

$$= 456820500000 \left(\frac{6561}{8}\right) = 456820500000(820.125)$$

Step 4: Simplify the result.

Performing the final arithmetic calculation:

$$F \approx \boxed{374,649,962,215,039.5000}$$

Thus, the total muscle force distributed in the region is approximately:

$$\mathbf{374,649,962,215,039.5000}$$

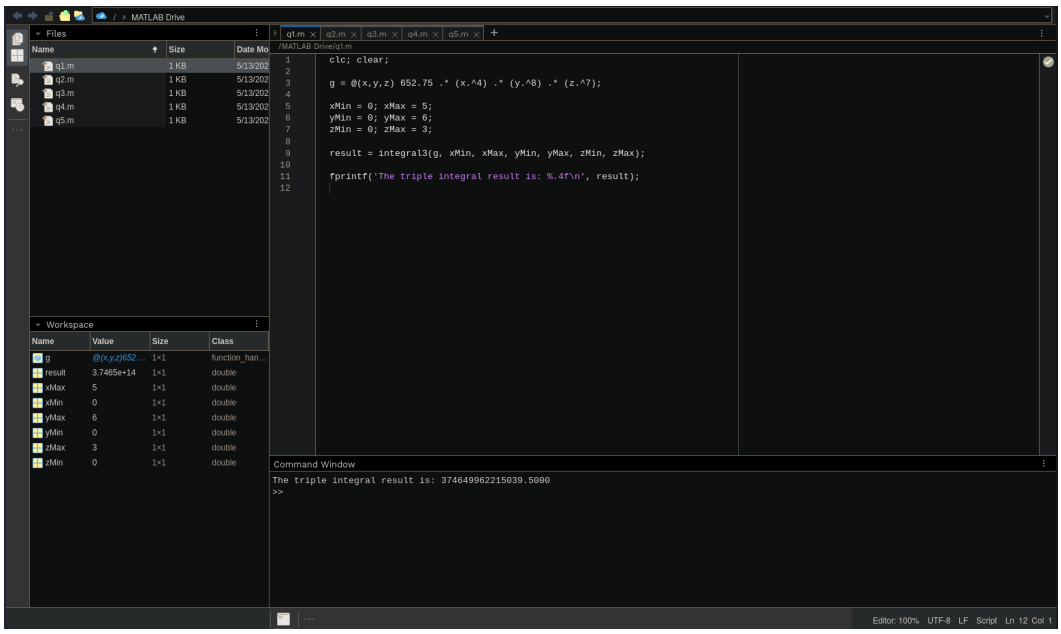
MATLAB Verification & Conclusion

MATLAB Code:

```
clc; clear;

g = @(x,y,z) 652.75 .* (x.^4) .* (y.^8) .* (z.^7);

xMin = 0; xMax = 5;
```

```
yMin = 0; yMax = 6;
zMin = 0; zMax = 3;

result = integral3(g, xMin, xMax, yMin, yMax, zMin, zMax);

fprintf('The triple integral result is: %.4f\n', result);
```

MATLAB Screenshot:



Conclusion:

The manual calculation for the total muscle force gives the **same** result as the MATLAB script.

Problem Statement Question 2

A biomechanics team models a spherical joint system with radius **7.5 units**, where the **joint force density** at a point (x, y, z) is described by:

$$I(x, y, z) = 600.17x^2 + 600.17y^2 + 600.17z^2$$

Compute the total joint force in the system using **spherical coordinates**.

Solution

The total joint force F is defined as the triple integral of the density function $I(x, y, z)$ over the spherical region E:

$$F = \iiint_E I(x, y, z)\, dV$$

First, we convert the density function and the region boundaries into spherical coordinates. The transformation is given by $x^2 + y^2 + z^2 = \rho^2$. Thus, the density function becomes:

$$I(\rho, \phi, \theta) = 600.17\rho^2$$

The volume element in spherical coordinates is $dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$. Since the region E is a full sphere of radius 7.5, the limits of integration are:

$$0 \le \rho \le 7.5, \quad 0 \le \phi \le \pi, \quad 0 \le \theta \le 2\pi$$

Applying the iterated integration process step-by-step:

$$F = \int_0^{2\pi} \int_0^{\pi} \int_0^{7.5} 600.17\rho^4 \sin\phi\, d\rho\, d\phi\, d\theta$$

Step 1: Integrate with respect to ρ.

Treating ϕ and θ as constants, we integrate the innermost function:

$$\int_0^{7.5} 600.17\rho^4 \sin\phi\, d\rho = 600.17 \sin\phi \left[\frac{\rho^5}{5}\right]_0^{7.5}$$

$$= 600.17 \sin\phi \left(\frac{151875}{32}\right) \approx 2848463.0859 \sin\phi$$

$$\Rightarrow F = \int_0^{2\pi} \int_0^{\pi} 2848463.0859 \sin\phi\, d\phi\, d\theta$$

Step 2: Integrate with respect to ϕ.

Next, we integrate the resulting expression with respect to ϕ:

$$\int_0^{\pi} 2848463.0859 \sin\phi\, d\phi = 2848463.0859 \Big[-\cos\phi\Big]_0^{\pi}$$

Substituting the bounds gives:

$$= 2848463.0859(-\cos\pi + \cos 0) = 5696926.1718$$

The integral expression simplifies to:

$$F = \int_0^{2\pi} 5696926.1718\, d\theta$$

Step 3: Integrate with respect to θ.

Finally, we evaluate the outermost integral with respect to θ:

$$\int_0^{2\pi} 5696926.1718\, d\theta = 5696926.1718 \Big[\theta\Big]_0^{2\pi} = 11393852.3437\pi$$

Step 4: Simplify the result.

Performing the final arithmetic calculation:

$$F \approx \boxed{35,794,842.8185}$$

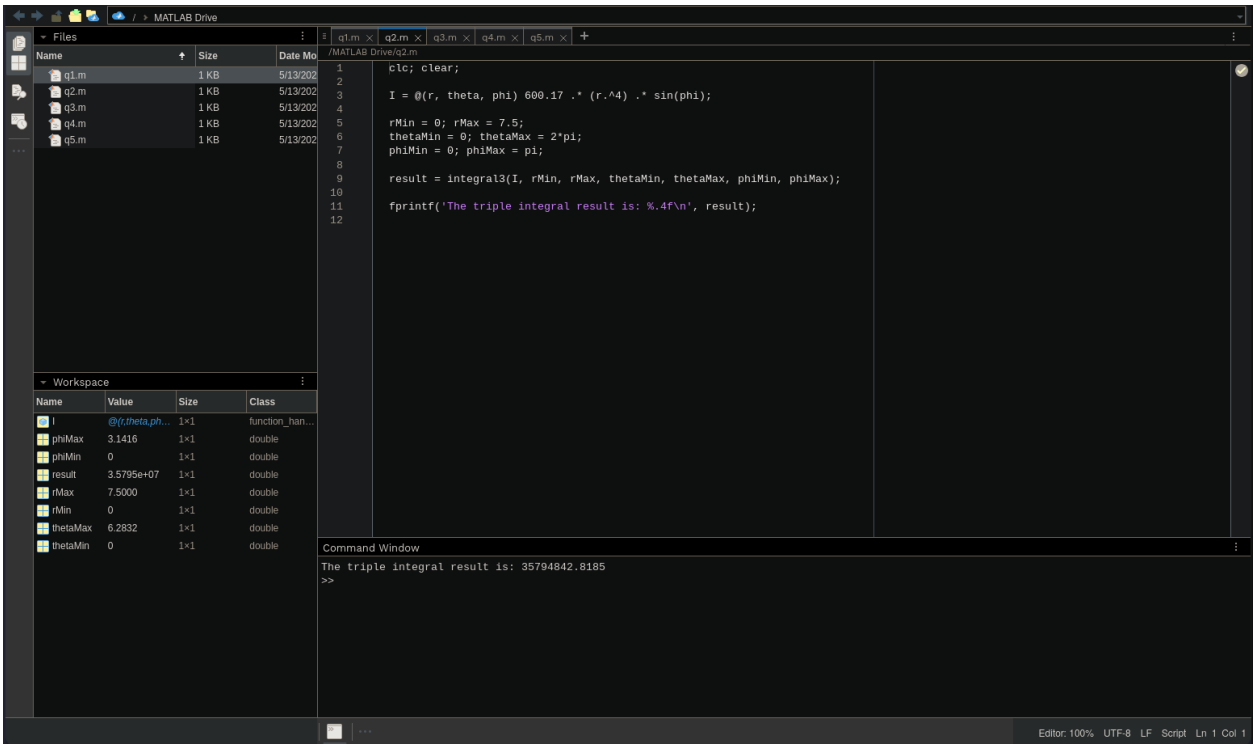
Thus, the total joint force distributed in the spherical system is exactly $\mathbf{35,794,842.8185}$.

MATLAB Verification & Conclusion

MATLAB Code:

```
clc; clear;

I = @(r, theta, phi) 600.17 .* (r.^4) .* sin(phi);

rMin = 0; rMax = 7.5;
thetaMin = 0; thetaMax = 2*pi;
phiMin = 0; phiMax = pi;

result = integral3(I, rMin, rMax, thetaMin, thetaMax, phiMin, phiMax);

fprintf('The triple integral result is: %.4f\n', result);
```

MATLAB Screenshot:



Conclusion:

By converting to spherical coordinates, the total joint force gives the **same** result as the MATLAB script.

Problem Statement Question 3

A cylindrical limb segment in a musculoskeletal model has a radius of **5.5 units** and a height of **9.5 units**. The **tendon stress density** at a point (x, y, z) within the segment is defined by:

$$P(x, y, z) = 354.62x^2 + 354.62y^2$$

Compute the total tendon stress in the segment using **cylindrical coordinates**.

Solution

The total tendon stress S is defined as the triple integral of the density function $P(x, y, z)$ over the cylindrical region:

$$S = \iiint_V P(x, y, z)\, dV$$

Converting the density function into cylindrical coordinates ($x^2 + y^2 = r^2$):

$$P(r, \theta, z) = 354.62r^2$$

The volume element in cylindrical coordinates is $dV = r\, dr\, d\theta\, dz$. The limits of integration for the cylinder are:

$$0 \leq r \leq 5.5, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq 9.5$$

Applying the iterated integration process step-by-step:

$$S = \int_0^{9.5} \int_0^{2\pi} \int_0^{5.5} 354.62r^3\, dr\, d\theta\, dz$$

Step 1: Integrate with respect to r.

$$\int_0^{5.5} 354.62r^3\, dr = 354.62 \left[\frac{r^4}{4}\right]_0^{5.5}$$

$$= 354.62 \left(\frac{14641}{64}\right) = 81124.8759$$

$$\Rightarrow S = \int_0^{9.5} \int_0^{2\pi} 81124.8759\, d\theta\, dz$$

Step 2: Integrate with respect to θ.

$$\int_0^{2\pi} 81124.8759\, d\theta = 81124.8759 \Big[\theta\Big]_0^{2\pi} = 162249.7518\pi$$

The integral expression simplifies to:

$$S = \int_0^{9.5} 162249.7518\pi\, dz$$

Step 3: Integrate with respect to z.

$$\int_0^{9.5} 162249.7518\pi \, dz = 162249.7518\pi[z]_0^{9.5} = 162249.7518\pi(9.5)$$

Step 4: Simplify the result.

$$S = 1540798.1171\pi \approx \boxed{4,842,364.3742}$$

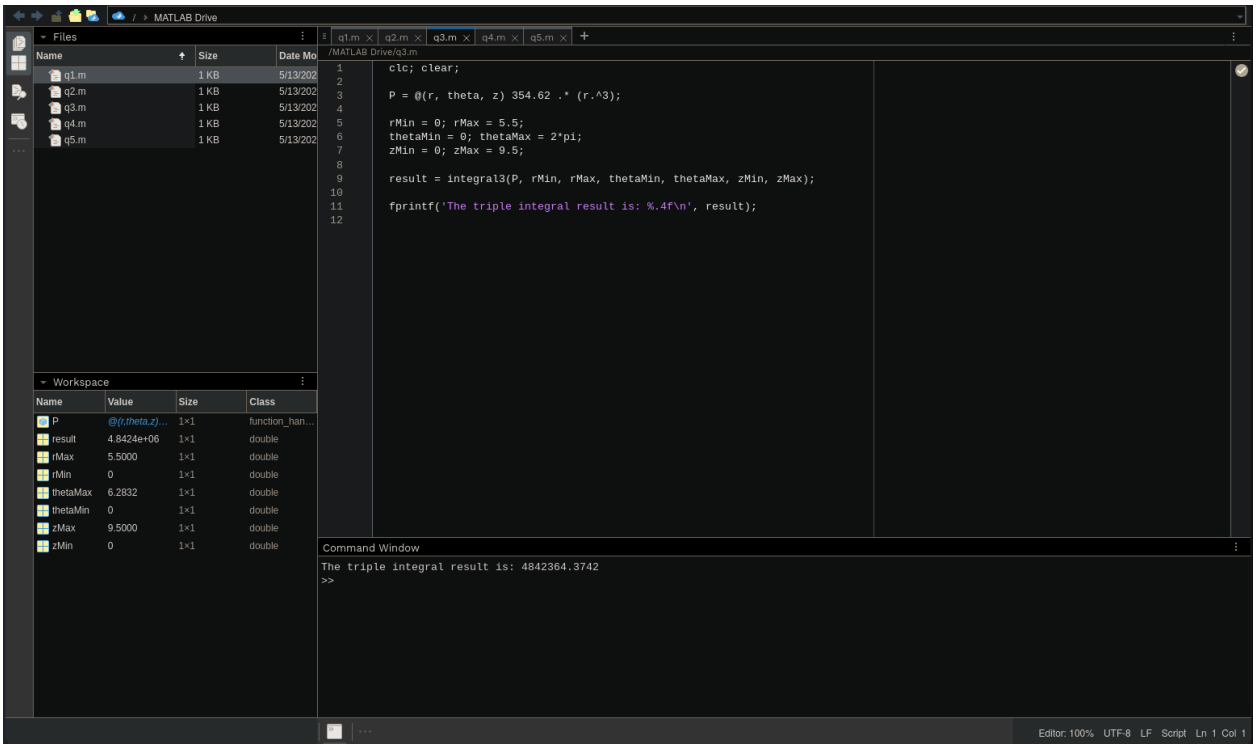
Thus, the total tendon stress distributed in the segment is exactly $\mathbf{4,842,364.3742}$.

MATLAB Verification & Conclusion

MATLAB Code:

```
clc; clear;

P = @(r, theta, z) 354.62 .* (r.^3);

rMin = 0; rMax = 5.5;
thetaMin = 0; thetaMax = 2*pi;
zMin = 0; zMax = 9.5;

result = integral3(P, rMin, rMax, thetaMin, thetaMax, zMin, zMax);

fprintf('The triple integral result is: %.4f\n', result);
```

MATLAB Screenshot:



Conclusion:

The total tendon stress calculated by using cylindrical coordinates gives the **same** result as the MATLAB script.

Problem Statement Question 4

In a multivariable musculoskeletal model, the **force output density function** for a limb system is:

$$P(x, y, z) = 901.17xyz$$

where x, y, z represent normalized inputs for muscle contraction, joint pressure, and bone displacement. A transformation to new coordinates is defined by:

$$x = \frac{u + 3v + w}{7}, \quad y = \frac{u - v + w}{7}, \quad z = \frac{u - v - 3w}{7}$$

Compute the **total musculoskeletal force output** over the region: $0 \leq u \leq 3$, $-1 \leq v \leq 6$, $-1 \leq w \leq 4$, using the **Jacobian transform**.

Solution

The total musculoskeletal force output F is calculated by evaluating the triple integral of the density function over the given region, translating to the uvw-space:

$$F = \iiint_G P(x(u,v,w), y(u,v,w), z(u,v,w))|J|\, du\, dv\, dw$$

First, we compute the Jacobian determinant J of the transformation:

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \left(\frac{1}{7}\right)^3 \begin{vmatrix} 1 & 3 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -3 \end{vmatrix}$$

$$= \frac{1}{343}\Big[1((-1)(-3) - (1)(-1))$$

$$-3((1)(-3) - (1)(1)) + 1((1)(-1) - (-1)(1))\Big]$$

$$= \frac{1}{343}\Big[1(4) - 3(-4) + 1(0)\Big] = \frac{16}{343} \approx 0.046647$$

The absolute value of the Jacobian is $|J| = \frac{16}{343}$. Let $A = u - v + w$. Substituting the transformation into the integral yields:

$$F = 0.122557 \int_{-1}^{4}\int_{-1}^{6}\int_{0}^{3} \left(A^3 + 4(v-w)A^2 - 16vwA\right)\, du\, dv\, dw$$

Applying the iterated integration process step-by-step:

Step 1: Integrate with respect to u.

$$\int_0^3 \left(A^3 + 4(v-w)A^2 - 16vwA\right)\, du = \left[\frac{A^4}{4} + \frac{4(v-w)A^3}{3} - 8vwA^2\right]_{u=0}^{u=3}$$

Expanding the bounds results in the following polynomial for v and w:

$$= 9v^3 + (21w - 22.5)v^2 - (21w^2 + 27w - 9)v - (9w^3 + 22.5w^2 + 9w - 20.25)$$

$$\Rightarrow F = 0.122557 \int_{-1}^{4}\int_{-1}^{6} \Big(9v^3 + (21w - 22.5)v^2$$

$$-(21w^2 + 27w - 9)v - (9w^3 + 22.5w^2 + 9w - 20.25)\Big)\, dv\, dw$$

Step 2: Integrate with respect to v.

Evaluating the polynomial with respect to v from -1 to 6:

$$\int_{-1}^{6} \Big(\ldots\Big)\, dv = -63w^3 - 525w^2 + 983.5w + 1585.5$$

The integral expression simplifies to:

$$F = 0.122557 \int_{-1}^{4} \left(-63w^3 - 525w^2 + 983.5w + 1585.5\right)\, dw$$

Step 3: Integrate with respect to w.

$$\begin{aligned} &\int_{-1}^{4} \left(-63w^3 - 525w^2 + 983.5w + 1585.5\right)\, dw \\ &= \left[\frac{-63w^4}{4} - 175w^3 + 491.75w^2 + 1585.5w\right]_{-1}^{4} \\ &= (-1022) - (-934.5) = -87.5 \end{aligned}$$

Step 4: Simplify the result.

$$F = 0.122557(-87.5) \approx \boxed{-10.7237}$$

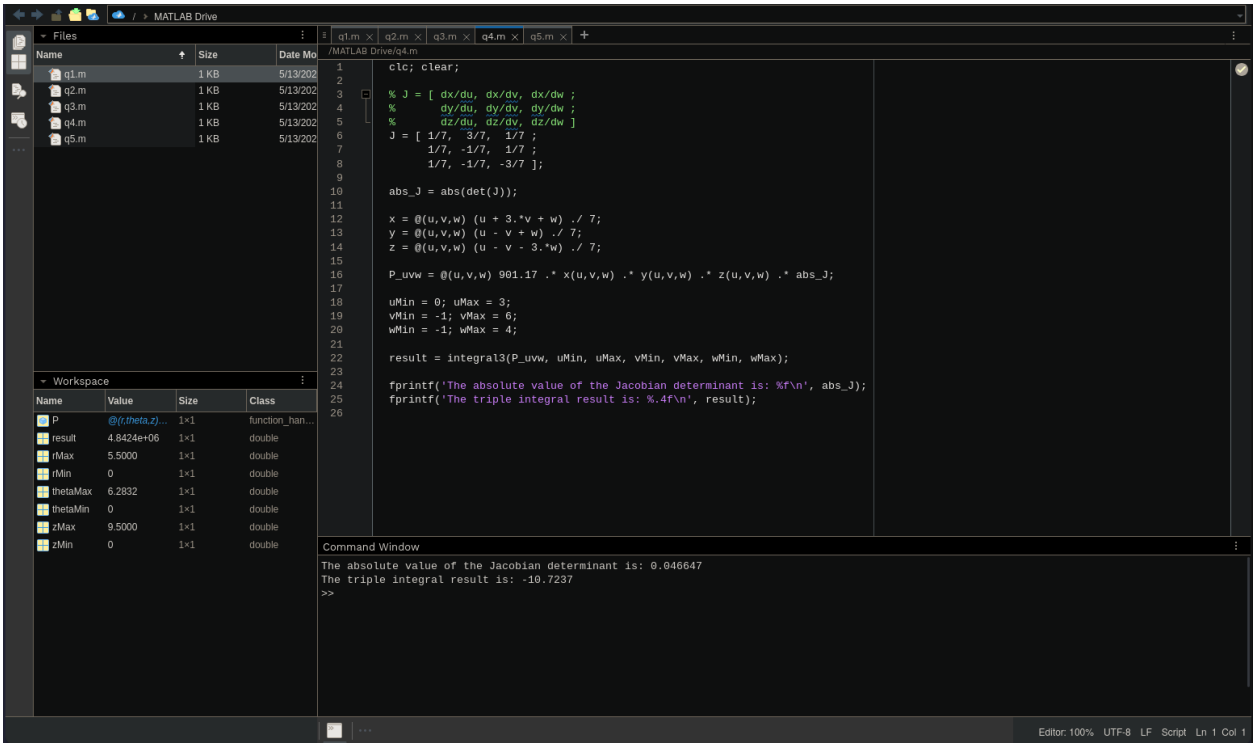
MATLAB Verification & Conclusion

MATLAB Code:

```
clc; clear;

% J = [ dx/du, dx/dv, dx/dw ;
%       dy/du, dy/dv, dy/dw ;
%       dz/du, dz/dv, dz/dw ]
J = [ 1/7,  3/7,  1/7 ;
      1/7, -1/7,  1/7 ;
      1/7, -1/7, -3/7 ];

```

```
abs_J = abs(det(J));

x = @(u,v,w) (u + 3.*v + w) ./ 7;
y = @(u,v,w) (u - v + w) ./ 7;
z = @(u,v,w) (u - v - 3.*w) ./ 7;

P_uvw = @(u,v,w) 901.17 .* x(u,v,w) .* y(u,v,w) .* z(u,v,w) .* abs_J;

uMin = 0; uMax = 3;
vMin = -1; vMax = 6;
wMin = -1; wMax = 4;

result = integral3(P_uvw, uMin, uMax, vMin, vMax, wMin, wMax);

fprintf('The absolute value of the Jacobian determinant is: %f\n', abs_J);
fprintf('The triple integral result is: %.4f\n', result);
```

MATLAB Screenshot:



Conclusion:

After applying the Jacobian transformation, the calculation for the force output gives the **same** result as the MATLAB script.

Problem Statement Question 5

The following triple integral represents the **force cost function** in a bounded musculoskeletal system with quadratic variation along the vertical axis:

$$\iiint_E (480.92x + 592.03y + z^2)dV$$

where E is the solid tetrahedron bounded by: $x = 0.1$, $y = 0.3$, $z = 0.4$, $x + y + z = 5$. Evaluate the **total force cost** within the system.

Solution

To set up the triple integral, we first determine the bounds of integration for the solid tetrahedron E:

- **Inner bounds (z):** Bounded below by $z = 0.4$ and above by the plane $z = 5 - x - y$.
- **Middle bounds (y):** $z_{\min} = 0.4 \Rightarrow y \in [0.3,\ 5 - 0.4 - x] \iff y \in [0.3,\ 4.6 - x]$
- **Outer bounds (x):** $y_{\min} = 0.3,\ z_{\min} = 0.4 \Rightarrow x \in [0.3,\ 5 - 0.3 - 0.4] \iff x \in [0.1,\ 4.3]$

The total force cost C is calculated by:

$$C = \int_{0.1}^{4.3} \int_{0.3}^{4.6-x} \int_{0.4}^{5-x-y} (480.92x + 592.03y + z^2)\, dz\, dy\, dx$$

Applying the iterated integration process step-by-step:

Step 1: Integrate with respect to z.

$$\int_{0.4}^{5-x-y} (480.92x + 592.03y + z^2)\, dz = \left[(480.92x + 592.03y)z + \frac{z^3}{3}\right]_{0.4}^{5-x-y}$$

$$= (480.92x + 592.03y)(4.6 - x - y) + \frac{(5 - x - y)^3}{3} - \frac{0.064}{3}$$

The integral expression becomes:

$$C = \int_{0.1}^{4.3} \int_{0.3}^{4.6-x} \left((480.92x + 592.03y)(4.6 - x - y) \right.$$

$$\left. + \frac{(5 - x - y)^3}{3} - \frac{0.064}{3} \right) dy\, dx$$

Step 2: Integrate with respect to y.

Evaluating the expanded polynomial with respect to y from 0.3 to $4.6 - x$ gives:

$$\int_{0.3}^{4.6-x} \Big(\ldots\Big)\, dy = 240.46(x^3 - 9.2x^2 + 21.16x) + \frac{592.03}{6}(4.6 - x)^3$$

$$+ \frac{(4.7 - x)^4}{12} + 144.276x^2 - 615.365x - 117.217$$

The integral expression simplifies to:

$$C = \int_{0.1}^{4.3} \left(240.46(x^3 - 9.2x^2 + 21.16x) + \frac{592.03}{6}(4.6 - x)^3 \right.$$

$$\left. + \frac{(4.7 - x)^4}{12} + 144.276x^2 - 615.365x - 117.217 \right) dx$$

Step 3: Integrate with respect to x.

$$\int_{0.1}^{4.3} \Big(\ldots\Big)\, dx = \left[240.46 \left(\frac{x^4}{4} - \frac{9.2x^3}{3} + 10.58x^2 \right) - \frac{592.03(4.6 - x)^4}{24} - \frac{(4.7 - x)^5}{60} \right.$$

$$\left. + 48.092x^3 - 307.6825x^2 - 117.217x \right]_{0.1}^{4.3}$$

Step 4: Simplify the result.

Evaluating the upper and lower bounds:

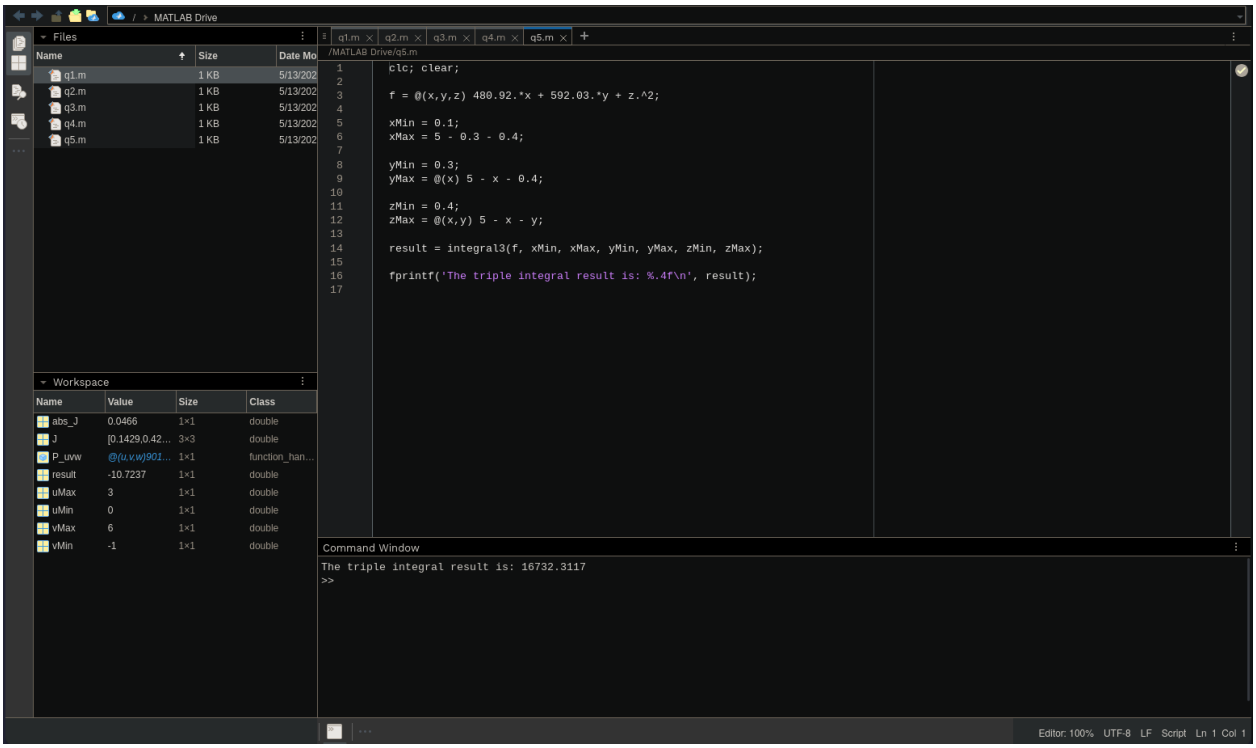
$$C \approx \boxed{16,732.3117}$$

MATLAB Verification & Conclusion

MATLAB Code:

```
clc; clear;

f = @(x,y,z) 480.92.*x + 592.03.*y + z.^2;

xMin = 0.1;
xMax = 5 - 0.3 - 0.4;

yMin = 0.3;
yMax = @(x) 5 - x - 0.4;

zMin = 0.4;
zMax = @(x,y) 5 - x - y;

result = integral3(f, xMin, xMax, yMin, yMax, zMin, zMax);

fprintf('The triple integral result is: %.4f\n', result);
```

MATLAB Screenshot:



Conclusion: The integration over the tetrahedral region results in the **same** value as the MATLAB script.